

NAVIN PANDIYAN

Lead AI Developer & ML Engineer

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70% automation of HTML review workflows via AI pipeline (SmarHTML) | **60% time savings** in daily research curation (NewsAI) | **28% efficiency gain** in debt collection across 800K+ accounts | **8+ apps** designed, built & deployed end-to-end

PROFESSIONAL SUMMARY

Lead AI Developer & ML Engineer with 7+ years delivering end-to-end AI-powered systems across healthcare, fintech, publishing, and enterprise domains. Expert in LangGraph-based agentic systems, serverless AWS architectures, production ML platforms, and LLM integration (OpenAI, Anthropic Claude). Proven track record of architecting, developing, and deploying complex AI pipelines from data ingestion to cloud infrastructure. Active Claude Code power user leveraging AI-native development workflows.

TECHNICAL SKILLS

AI/LLM: LangGraph, LangChain, RAG (pgvector, FAISS), Multi-Agent Systems, MCP, Prompt Engineering, deepagents

LLM Providers: Claude (Anthropic), GPT-4o/o1 (OpenAI), Gemini (Google), DeepSeek, Ollama

ML/DS: XGBoost, CatBoost, LightGBM, PyTorch, Transformers, PEFT/LoRA/QLoRA, SHAP, scikit-learn, spaCy

Backend: Python, FastAPI (async), SQLAlchemy 2.0, Redis, WebSockets, Socket.IO

Data Platforms: Databricks, Spark SQL, Snowflake, MLflow, Alembic, Feature Stores

Cloud/AWS: Lambda (10GB), EC2, ECS Fargate, S3, RDS, Cognito, CloudFront, SQS, SAM/CloudFormation

Databases: PostgreSQL (+pgvector), MongoDB, Neo4j, Azure Cosmos DB, MySQL, DynamoDB

CV/NLP: OpenCV (ORB/SIFT), Azure OpenAI Vision, Tesseract OCR, Playwright, NLTK

DevOps: Docker, Docker Compose, Nginx, GitHub Actions, ECR, CI/CD

AI Tools: Claude Code CLI (daily driver), MCP servers, YAML prompt templates, agentic TDD workflows

PROFESSIONAL EXPERIENCE

Lead AI Developer August 2024 - Present

Digit88, Bangalore

- Designed & deployed SmarHTML, a multi-stage AWS Lambda pipeline (SQS-orchestrated) that automates PDF-to-HTML review workflows using computer vision (ORB/SIFT + dhash) and LangGraph+MCP agents, reducing developer time by 70% (1-2 hrs to ~25 min per asset)
- Built NewsAI, a LangGraph-orchestrated agentic newsletter platform with multi-provider LLM support, replacing 6 hours of daily analyst work with 10-minute automated runs, processing 200-300 articles in 15-20 min
- Developed AutoDoc AI, a LangGraph-based codebase intelligence platform that analyzes repositories to surface architectural insights, vulnerabilities, and improvement areas - with a roadmap toward agentic developer workflows (automated refactoring, PR generation, Jira integration) and a validation/audit layer

Senior Data Science Analyst II July 2023 - August 2024

Dun & Bradstreet, Chennai

- Led the CollectorAssist ML Platform with multi-model system (ITP, PRE, PTP, RTtC) on Databricks, increasing debt collection efficiency by 28% and agent success rates by 36% across 800K+ accounts
- Built SuperModel training framework across 5 business units with automated daily scoring, model versioning (UUID-based), and probability calibration (CalibratedClassifierCV)
- Implemented T-Learner causal inference models; automated MLOps pipelines with MLflow and Databricks Job Schedulers
- Built explainable AI dashboards (SHAP/LIME) for transparent insights and client adoption

Data Scientist - Consultant December 2022 - August 2024

Remote (Freelance)

- Played a crucial role as Lead AI Developer & Backend Engineer on CareInsight AI, driving the healthcare analytics product to client go-live deadlines by building the LLM-based KPI extraction pipeline, full i18n translation layer, and automated form fillers for patient data using structured document retrieval
- Delivered cloud cost optimization and anomaly detection frameworks, reducing unnecessary cloud expenditure by 20% through predictive analytics
- Developed MedAssist AI, a voice-enabled hospital discovery agent using LangGraph + RAG + Whisper STT with QLoRA fine-tuning

Senior Data Scientist / Data Scientist January 2019 - December 2022

Pitney Bowes, Pune

- Developed NLP pipelines for document classification achieving 94% accuracy across 100K+ records; built reusable feature engineering toolkits reducing preprocessing time by 35%
- Led forecasting, object detection, and anomaly detection projects for shipping/eCommerce logistics including customs optimization, HS code classification, and warehouse volume prediction

KEY PROJECTS (Designed, Built & Deployed End-to-End)

Bharat2Business *(FastAPI, LangGraph RAG, AWS, pgvector)*

- B2B MSME platform with multi-node LangGraph RAG agent, automated KB pipeline (Crawl4AI + Docling + BGE-M3), 14 API routers, fully deployed on AWS with SAM/CloudFormation IaC

SmartHTML *(5 AWS Lambdas, SQS, OpenCV, Azure Vision, LangGraph+MCP, Playwright)*

- Multi-stage serverless pipeline automating PDF annotation-to-HTML updates; two-tier vision matching (dhash + ORB/SIFT), Playwright state discovery, LangGraph+MCP edit agents; 70% automation rate within 8 min

NewsAI *(LangGraph, multi-provider LLM, Socket.IO, Neo4j, Lambda)*

- Agentic newsletter curation platform replacing 6 hrs/day of analyst work; guided/auto modes, LLM article ranking, parallel summarization, daily newsletters in under 10 min

AutoDoc AI *(LangGraph, deepagents, PostgreSQL checkpoints, Mermaid, Cognito, S3)*

- AI codebase intelligence platform that analyzes repos, surfaces pitfalls, and generates discovery documents; roadmap includes agentic developer workflows (MCP-based PR generation) and architecture validation/audit layer

EDUCATION

M.Tech in Data Science & Engineering, BITS Pilani (2021-2023)

B.Tech, SSN College of Engineering (2013-2017)

PG Diploma in Data Science & Analytics, Imarticus Learning (2018)

CERTIFICATIONS
